

Investment Memo - Lemurian Labs

Track A: AI Infrastructure

1. Executive Summary

What the company does:

Lemurian Labs is building a hardware-agnostic runtime and compilation software layer for AI workloads, aiming to let developers “write once, run anywhere” across heterogeneous compute environments (GPUs, CPUs, and specialized accelerators). Its software technology stack, Tachyon, seeks to reduce hardware differences while preserving near-native performance of AI models¹.

Who pays for Tachyon?

The primary economic buyer is the head of platform engineering or AI infrastructure at organizations running production AI systems, while end users are ML engineers and systems teams responsible for deployment and optimization. Buyers are motivated by rising NVIDIA GPU costs, underutilization of expensive accelerators, and the growing operational burden of maintaining vendor-specific optimization paths (e.g., NVIDIA's CUDA-locked workloads²³). As AI workloads expand beyond research into large-scale inference and embedded product features, the ability to run models efficiently across CPUs, GPUs, and heterogeneous infrastructure becomes a cost and velocity constraint rather than a technical preference⁴.

Why is Tachyon useful?

Frameworks like CUDA are essential to running high-performance AI workloads, but they lock developers into NVIDIA GPUs because CUDA is a proprietary programming model tied exclusively to NVIDIA hardware⁵. Code written as CUDA kernels cannot natively run on other GPUs or on CPUs without being rewritten. Lemurian Labs' Tachyon stack aims to avoid this lock-in by compiling AI models into a hardware-neutral representation and then generating optimized, hardware-specific execution paths for different targets, including NVIDIA GPUs and CPUs. Rather than translating CUDA kernels, Tachyon seeks to replace the need for hardware-specific kernels altogether by internalizing hardware differences inside its compiler and runtime layer⁶.

Why now?

AI used to run mostly on NVIDIA GPUs, so companies built their systems around one type of hardware. Today, AI runs across a mix of machines—NVIDIA GPUs, CPUs, different cloud servers, and on-prem systems. This makes it hard and expensive to move AI models between environments, because code is often written and optimized for one specific type of hardware⁷.

¹ <https://www.lemurianlabs.com>

² <https://www.ft.com/content/e56dd467-7a4d-437d-b7f8-f0ef2320b1ee>

³ <https://docs.nvidia.com/cuda/cuda-c-programming-guide/index.html>

⁴ <https://aiindex.stanford.edu>

⁵ <https://developer.nvidia.com/cuda-zone>

⁶ <https://tvm.apache.org/docs/>

⁷ <https://www.mckinsey.com/capabilities/quantumblack/our-insights>

As NVIDIA GPU costs have risen and hardware has become more fragmented, the ability to run the same AI model on different machines without rewriting it has become a real cost and speed problem, not just a technical preference². Lemurian Labs' Tachyon stack addresses this by handling hardware differences in software, allowing teams to deploy models across heterogeneous infrastructure more easily.

Top killer risk:

If heterogeneous compute is such an obvious problem, why hasn't a big company already solved it? Is this an unsolved problem because it's early — or because it's hard? The absence of a dominant, vendor-neutral solution for heterogeneous AI execution suggests the problem is constrained by technical difficulty rather than lack of demand⁸. Lemurian's success therefore hinges on the quality of its compiler and runtime stack. If the platform cannot reliably deliver near-native performance across heterogeneous hardware without bespoke tuning, sophisticated teams will continue to tolerate vendor lock-in and higher compute costs, collapsing the core value proposition.

2. Problem, Buyer, and Wedge

The Problem and Solution:

- Problem faced by industry: Hardware lock-in, rising NVIDIA GPU dependence, and non-scalable optimization effort.
- Solution: Lemurian Labs is attempting to address this constraint by internalizing hardware-specific complexity inside a compiler and runtime layer, rather than leaving it to application and infrastructure teams.

Buyer and User:

- Economic buyer: Platform / infrastructure leadership (CTO, Head of AI Platform)
- End user: ML engineers and systems engineers deploying and optimizing models
- Budget owner: Centralized infrastructure or AI platform budget, not individual application teams

This separation increases adoption friction but also concentrates purchasing power if Lemurian becomes infrastructure-critical.

Adoption into real workflows:

Lemurian is most likely adopted first in lower-risk parts of the AI workflow, such as inference, experimentation, or cost-sensitive workloads that do not require peak GPU performance. These use cases allow teams to test the platform without putting critical systems at risk. If Lemurian demonstrates reliable performance and operational benefits, usage can expand over time to more important production workloads.

⁸ <https://docs.cloud.google.com/tpu/docs/pytorch-xla-performance-profiling-tpu-vm>

3. Differentiation and Defensibility

Competitive advantage:

- Unique hardware-agnostic software stack: Tachyon is a compiler/runtime stack that meaningfully reduces hardware-specific optimization effort while preserving performance.
- Workflow lock-in: Once Tachyon is integrated into deployment pipelines, switching costs rise due to tooling, monitoring, and CI/CD dependencies.
- Ecosystem leverage: Tachyon has potential to become a neutral layer across clouds, accelerators, and different hardware – positioning Lemurian as a continuous infrastructure glue (think consistent cashflows) rather than a one point solution.

Where the moat might be overstated:

- Hardware-agnostic compilation is an active area of academic and open-source innovation, notably Apache TVM⁹, XLA¹⁰, and MLIR¹¹.
- Large incumbents (cloud providers, chip vendors) may tolerate inefficiency in portability in exchange for ecosystem lock-in, limiting Lemurian's leverage with the largest buyers¹²¹³.

In AI infrastructure, technical advantages tend to diffuse quickly through open-source and incumbent platforms such as established cloud providers and hardware vendors. As a result, defensibility for Lemurian Labs is unlikely to come from compiler techniques alone, but from Tachyon's execution quality, workflow integration, and adoption depth. For Lemurian, durable advantage is more likely to emerge from being embedded in production deployment pipelines and trusted for reliability, rather than from pure technical novelty.

While open-source compilers demonstrate technical feasibility, Lemurian's bet is that most organizations will pay for a productized, reliable abstraction rather than operate compiler infrastructure themselves.

4. Market sanity check

Step 1: Defining the relevant universe

The relevant market is organizations running AI workloads in production at meaningful scale, not hobbyist users or research-only teams. This includes mid-to-large enterprises and AI-native companies deploying models for inference, personalization, fraud detection, search, forecasting, or operational decision-making.

⁹ <https://tvm.apache.org>

¹⁰ <https://www.tensorflow.org/xla>

¹¹ <https://mlir.llvm.org>

¹² <https://aws.amazon.com/machine-learning/inferentia/>

¹³ <https://cloud.google.com/tpu>

Step 2: Estimating the number of such organizations

There are several hundred thousand mid-to-large enterprises globally, but only a subset operate AI systems in production. Assuming that roughly 20–25% of these organizations have deployed AI at scale implies on the order of tens of thousands of organizations worldwide. For a back-of-the-envelope estimate, this analysis assumes approximately ~50,000 organizations running production AI workloads at meaningful scale.

Step 3: Defining the relevant spend category

Lemurian is not targeting total compute spend, but AI infrastructure and deployment tooling spend, which includes platform software, optimization layers, deployment tooling, and operational support. This is a budget typically owned by centralized infrastructure or AI platform teams.

Step 4: Estimating average annual spend per organization

For organizations running production AI systems, annual spend on infrastructure tooling is conservatively estimated at \$50k–\$200k per year, depending on company size, workload complexity, and deployment footprint. This range is modest relative to overall compute spend and consistent with pricing for other infrastructure-level software.

Step 5: Deriving the implied market size

Multiplying ~50,000 organizations by \$50k–\$200k in annual infrastructure tooling spend yields an implied serviceable market of approximately \$2.5B–\$10B annually, before accounting for TAM CAGR components like expansion from increased AI adoption, new workloads, or deeper platform penetration.

Step 6: Interpreting the result

This estimate is intended to validate order of magnitude, not produce a precise TAM. Even under conservative assumptions, the market supports a multi-billion-dollar opportunity, which is sufficient to justify venture-scale outcomes if Lemurian can capture a meaningful share.

Cautionary note about TAM estimate:

This estimate is intentionally conservative. The most sensitive assumptions are the number of organizations running AI workloads in production, and the number of different teams running team-independent workloads per organization. As AI adoption expands beyond centralized data science groups into product, operations, and internal tooling teams, infrastructure spend can grow materially within existing customers even if per-team budgets remain modest.

5. Business model and unit-economics

Value capture:

Lemurian captures value by reducing total AI infrastructure cost through enabling cost-appropriate hardware usage and minimizing ongoing hardware-specific engineering effort.

Economic anchor:

Many inference and batch workloads are run on expensive NVIDIA GPUs despite not requiring

peak performance; shifting even a portion to optimized CPU or alternative cheaper hardware can generate meaningful compute savings while reducing optimization overhead.

Pricing model:

Given its position in the infrastructure stack, Lemurian is likely to price based on deployment footprint or usage (e.g., per deployment, per cluster, or usage-based, similar to infrastructure and cloud software). Precise pricing assumptions are difficult to ground at this stage, as Tachyon appears to still be in active development with no publicly disclosed contract values. Unlike CUDA, which is free to developers and monetized indirectly through NVIDIA hardware, Lemurian's value capture would need to come directly from software economics tied to realized cost savings and operational leverage¹⁴. This makes any direct pricing comparisons difficult.

Adoption wedge:

Initial adoption is likely in lower-risk, cost-sensitive workloads where performance tradeoffs are acceptable, allowing teams to validate reliability without putting critical systems at risk.

Expansion dynamics:

As performance confidence grows, usage can expand across additional workloads and teams within an organization, tying revenue growth to workload volume and deployment breadth rather than simply the number of users.

6. Competitors and substitutes

Direct / adjacent competitors:

- Apache TVM⁹
 - What it is: Apache TVM is an open-source machine learning compiler stack that lowers models into hardware-specific optimized code across CPUs, GPUs, and accelerators.
 - Where it's strong:
 - Technically powerful and flexible
 - Actively used in research and by sophisticated infra teams
 - Supports many hardware backends
 - Why it doesn't fully solve the problem: TVM is not a turnkey product. It requires deep compiler expertise, extensive tuning, and ongoing maintenance to achieve strong performance. As a result, it is best suited for organizations with large infra teams rather than the broader market Lemurian targets.
 - Why this matters for Lemurian: TVM validates the technical approach, but also highlights that abstraction alone is insufficient—usability, reliability, and operational integration determine adoption. Lemurian's differentiation is productizing this capability for teams that cannot afford compiler-level complexity.
- XLA¹⁰
 - What it is: XLA (Accelerated Linear Algebra) is a compiler developed by Google to optimize ML workloads, primarily within TensorFlow and JAX.

¹⁴ <https://developer.nvidia.com/cuda-toolkit>

- Where it's strong:
 - Deeply optimized for Google's hardware (TPUs)
 - Mature, battle-tested at massive scale
 - Strong performance for supported workloads
- Why it doesn't fully solve the problem: XLA is tightly coupled to Google's ecosystem and design assumptions. It is not vendor-neutral and offers limited portability across non-Google hardware and environments.
- Why this matters for Lemurian: XLA demonstrates that near-native performance abstraction is possible—but only within a vertically integrated stack. Lemurian's challenge (and opportunity) is delivering similar abstraction without owning the hardware or cloud.
- ONNX¹¹
 - What it is: ONNX is a standardized format for representing ML models so they can be moved between frameworks and runtimes.
 - Where it's strong:
 - Widely adopted interoperability standard
 - Enables model portability across tools and vendors
 - Low friction to adopt
 - Why it doesn't fully solve the problem: ONNX stops at model representation. It does not handle execution, scheduling, memory management, or performance optimization. Teams still need a runtime or compiler stack underneath.
 - Why this matters for Lemurian: ONNX is complementary, not competitive. Lemurian could ingest ONNX models, but its value lies in execution and performance, not interchange.

Status quo substitute:

Internal, hardware-specific optimization pipelines maintained by ML infrastructure teams.

Most credible competitive attack vector:

Cloud providers and hardware vendors already offer tools that reduce some of the complexity of running AI, such as managed ML platforms, deployment pipelines, and Software Development Kits¹²¹³ tailored to their own hardware. These tools make it easier for teams to train and deploy models without managing all infrastructure details themselves, but they are typically proprietary and ecosystem-specific, not truly hardware-agnostic.

While these offerings do not fully solve the problem Lemurian targets, they are often “good enough” for many customers. Over time, cloud providers and hardware vendors could expand these built-in tools further, reducing the need for an independent, vendor-neutral solution. As a result, Lemurian's competitive risk is not hypothetical: incumbents already address parts of the problem today and could deepen their abstractions if customer demand warrants it.

7. Risks and de-risking measures

Risks	Evidence to De-Risk	Diligence Approach
<u>Performance abstraction fails at scale</u> If Lemurian cannot deliver near-native performance across heterogeneous hardware without bespoke tuning, advanced teams will reject the platform for production workloads.	Demonstrated benchmark parity (or acceptable tradeoffs) across CPUs, GPUs, and mixed environments.	Review third-party benchmarks; compare performance vs native CUDA and optimized CPU baselines.
<u>Adoption stalls beyond pilots</u> If Lemurian remains limited to experimentation or non-critical workloads, it may fail to become core infrastructure.	Evidence of production deployments beyond pilots, with repeat usage.	Analyze customer cohorts; assess pilot-to-production conversion rates.
<u>Incumbent tools are “good enough”</u> If existing tools (e.g., TVM, XLA, cloud-native stacks) satisfy customer needs with less friction, Lemurian may struggle to displace them.	Clear differentiation in usability, performance consistency, or operational simplicity.	Review technical roadmap; compare required effort to achieve similar outcomes with incumbents.
<u>Limited pricing power</u> If customers view Lemurian as an optimization layer rather than core infrastructure, willingness to pay may be capped.	Customers anchoring value to compute or engineering cost savings rather than feature-level pricing.	Hypothetical customer interviews; assess budget ownership and procurement behavior.

<u>Vendor or ecosystem dependence emerges</u> If Lemurian becomes dependent on specific hardware vendors or cloud platforms, its neutrality and strategic value diminish.	Demonstrated support across multiple hardware vendors and deployment environments.	Review partner strategy; assess contractual or technical dependencies.
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8. Diligence asks

8 Management Questions:

- Where does Tachyon underperform native stacks today, and why?
- What workloads are explicitly out of scope?
- How often do customers fall back to hardware-specific tuning?
- What internal expertise is required to deploy Lemurian successfully?
- How do you benchmark against TVM or XLA internally?
- What customer segment shows the highest expansion behavior?
- How do you defend neutrality against cloud vendor pressure?
- What must be true technically for Lemurian to win long-term?
- On their website, this is where Lemurian Labs currently stand, “We're building the hardware-agnostic software infrastructure that makes AI accessible to everyone—fast, affordable, and scalable. Not just better tools. The foundation for what comes next. If we succeed, we enable the future.” The “if we succeed” part leads me to believe Lemurian Labs is operating more as a research lab than a startup with a deliverable product. This raises my skepticism on the ability to build Tachyon, so it would be good to have a timeline for when Tachyon is planned to release.

5 Datapoints Requested:

- Lemurian Labs’ pricing strategy for Tachyon
- Performance benchmarks vs native stacks across hardware
- Deployment success rate from pilot → production
- Retention by workload type
- Average compute cost savings realized
- Sales cycle length by customer size

9. Recommendation: Need more data + Proceed to diligence (because this is an interesting opportunity)

Lemurian Labs is addressing a real and growing constraint in AI infrastructure: rising dependence on NVIDIA GPUs, increasing hardware fragmentation, and the operational burden of hardware-specific optimization. The market opportunity is directionally attractive, and the company's thesis is well aligned with where production AI workloads are heading.

However, the investment case ultimately hinges on whether Tachyon can reliably deliver near-native performance across heterogeneous hardware without requiring bespoke tuning. This technical bar is high and has historically limited vendor-neutral solutions in this space. Before forming conviction, further diligence is required on third-party performance benchmarks, evidence of production deployments beyond pilots, and a clear timeline to product readiness.

If Lemurian can demonstrate that its software stack holds in real production environments, the opportunity could justify deeper investment consideration. Absent this proof, the core value proposition risks breaking down in practice despite strong market tailwinds.